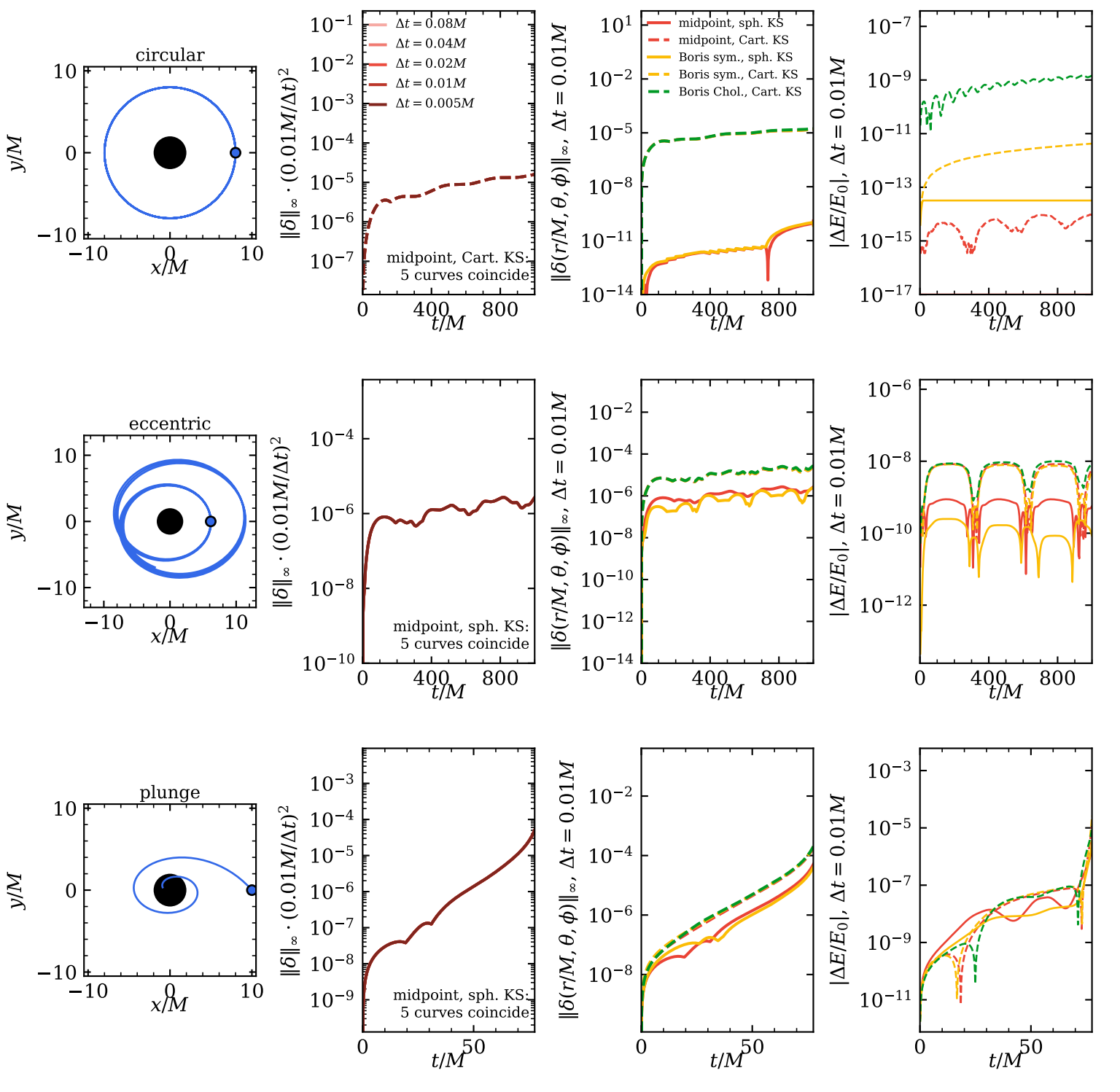

circular
eccentric
plunge
y/M
x/M
t/M
Δt = 0.08M
Δt = 0.04M
Δt = 0.02M
Δt = 0.01M
Δt = 0.005M
midpoint, Cart. KS:
5 curves coincide
midpoint, sph. KS:
5 curves coincide
midpoint, sph. KS
midpoint, Cart. KS
Boris sym., sph. KS
Boris sym., Cart. KS
Boris Chol., Cart. KS
‖δ‖∞ · (0.01M/Δt)²
‖δ(r/M, θ, φ)‖∞, Δt = 0.01M
|ΔE/E0|, Δt = 0.01M